Examiner reports

Q1.

Most candidates made a confident start to this question. In part (a)(i), they usually standardised 60 using 56 and 2.5 though a very small minority used $\sqrt{2.5}$ or 2.5^2 in the denominator. However rather more candidates standardised 59 (0 marks) or 59.5 (1 mark). Part (a)(ii) was also answered well with many candidates scoring full marks. When marks were lost, the reason was usually for not dealing with **Φ**(– 2.4) as 1 – **Φ**(2.4); a difficulty that may not have arisen had such candidates indicated the required area on a sketch of a normal curve.

Very few candidates had correct answers to part (a)(iii). Apparently few candidates were aware that $P(X = x) = 0$ when x is continuous and so often attempted $P(54.5 < X < 55.5)$ despite there been only 1 mark available. A majority of candidates scored at least 3 of the 4 marks available in part (b); this lost mark was through using z = + 2.0537. A number of candidates lost at least 3 marks for equating $\frac{100 - \mu}{3.4}$ to the probability for z = 0.98 from Table 3 of the supplied booklet.

Q2.

Notation and presentation of method were in need of improvement on many scripts although many candidates scored high marks for correct answers. In part (a), almost all candidates knew how to standardise 250 using 200 and 25 and so obtained a correct answer. A minority of candidates used 249, 249.5 or 249.9, employed a divisor of 625 or 5, or, having found z = 1.8 correctly, chose to use 1.79 in Table 3. Fewer candidates were completely successful in part (a)(ii), usually through not making the area change, **Φ**(– 0.2) = 1 – **Φ**(0.2). In part (b)(i), many candidates made the mistake of using 0.5244 or 1 – 0.5244 rather than – 0.5244, whilst a minority used 0.70 as a z-value in Table 3. More candidates answered part (b)(ii) correctly since the z-value needed was positive, but a minority read 'medium' as 'median' and so used z = 0. There was a surprisingly large proportion of correct answers to part (c) often from candidates who had answered part (b) incorrectly. One general point of note for the future: too many candidates lose valuable accuracy marks through finding approximate z-values from Table 3 rather then more exact values from Table 4.

Q3.

This opening question allowed almost all candidates to score some marks since standardising was almost always of the correct form. In part (a) a significant number of candidates obtained the probability of 'less than 60 cm' and then found that they had an identical answer for part (b). Only a small proportion of such candidates corrected their answers to part (a) to zero. It was pleasing to see the number of candidates capable of scoring full marks in part (c). On the other hand, however, it was somewhat of a shock to see the number of weak candidates unable to find 5% of 48; sometimes this inability appeared disguised as 48 ± 0.05 or 48 ± (z-value).

Q4.

All but the weakest candidates were able to achieve the four marks in part (a). Fewer candidates tried introducing a continuity correction or confused σ with σ^2. However, confusion as to whether $\Phi(z)$ or $1 - \Phi(z)$ was required was a significant problem. In part (b), many candidates equated $\frac{10 - 10.25}{\sigma}$ to + 2.0537 and then either left their answer as negative or 'lost' the minus sign. Few candidates had equations involving 0.98, 0.02 or 1 – 2.0537.

Q5.

In this question there were noticeably fewer candidates than on previous papers attempting to introduce continuity corrections or to use √0.5 or 0.5^2 rather than 0.5.

Most candidates scored full marks in part (a)(i) although a few gave Φ(1.6) as their answer. Again, in part(a)(ii), full marks were the norm with the most noticeable error being that of evaluating Φ(1.4) – Φ(0.6).

In part (b), most candidates realised that $\frac{w - 25.8}{0.5}$ had to be equated to some value from tables. Many candidates then lost at least 1 mark for using 0.67 or 0.68 from Table 3, rather than 0.6745 from Table 4, and/or not including a minus sign. Some candidates lost at least 2 of the 4 marks available for using $1 - z_{0.75}$, Φ(0.75) or 1 – Φ(0.75).

Q6.

Most candidates scored full marks in part (a)(i) and it was pleasing to see a reduction in both the use of a continuity correction and the confusion between σ and σ^2 . Similarly, answers to part (a)(ii) showed a marked improvement with fewer candidates using Φ(0.9) or $1 - z_{0.9}$ rather than $z_{0.9}$. However, many candidates obtained an estimate of the latter as 1.285 or 1.29 from Table 3 rather than Table 4 of the formulae booklet. Such candidates lost at least the final accuracy mark. In part (b), weaker candidates often equated $\frac{x - \mu}{\sigma}$ to 0.14 and 0.03 and so scored at most one mark. Average candidates attempted to find z-values but often used Table 3 and so were penalised for inaccuracy. Such candidates rarely made much progress in solving their two equations. Better candidates, who obtained values of –1.08(03) and +1.88(08) from Table 4, often went on to solve their two equations and score full marks.

Q7.

Almost without exception, candidates scored well on this question, with full marks the most common score. In part (a), candidates usually obtained 0.64499 by a correct method though, on rare occasions, some valid or invalid adjustments were made retrospectively to obtain the quoted answer. The number of completely correct answers to part (b) was pleasantly surprising. Incorrect attempts usually involved $\frac{6}{10}$ × 0.645, $X \sim B(10, 0.6)$ or $\overline{X} \sim N\left(208.5, \frac{2.5^2}{6}\right)$.

Q8.

Many candidates, particularly the more able ones, were able to score high marks on this question. In part (a)(i), weaker candidates were unable to cope correctly with finding P(Z > –1.4), despite sketches showing that a probability greater than 0.5 was required. Apparent failure to understand 'one bar in five hundred' and/or find a corresponding, or indeed any, z-value were the main weaknesses in answers to part (a) (ii). Those candidates who translated to 0.002 with a z-value of 2.88 invariably went on to score the 5 marks available. There was a pleasing proportion of correct, or almost correct, answers to part (b). However, many candidates appeared to struggle with finding correct z-values for 1.5% and 1.9%, in terms of magnitude and/or sign; some were clearly unaware of Table 4 in the formulae booklet. Many candidates made hard work of solving the two simultaneous equations with some abandoning their inefficient or incorrect attempts, which were often not helped by at least one z-value having an incorrect sign.

Q9.

Part (a) was well answered. The most common mistake in part (b) was to use the wrong tail of the distribution.

Q10.

There were few problems with this question.

Q11.

This was not a high scoring question. Many candidates failed to realise that in part (a)(ii) they needed to calculate the probability of consumption being between 20 litres and 40 litres.

There were some good answers to part (b). Part (c) was not well answered. Although most candidates successfully found the z-values, the equations often contained incorrect signs which lead to nonsensical answers.

Q12.

This question was mostly answered well.

Q13.

This question caused few problems although, inevitably, some candidates attached the wrong sign to their z-values in parts (a)(ii) and (b).

Q14.

Most candidates were well prepared for part (a). Most candidates found part (b) to be the most demanding part of this question but some candidates were obviously well prepared and gained full marks with ease.

Q15.

This was a standard question and produced some excellent answers. Rather more candidates than expected were unable to make headway past part (a).

Q16.

Part (a) was well answered, but many candidates were unable to decide what was required in part (b).

Q17.

Part (a) posed few problems but a wide variety of answers were given in part (b)(i). A number of candidates rounded the correct answer to part (b)(i) to 3 significant figures. In view of the rubric this was accepted but those who used this rounded figure in part (b)(ii) lost accuracy marks.

A few candidates used spurious “continuity corrections”. This should be discouraged, although it is usually treated sympathetically. On this occasion, weights were given to one decimal place and so using 124.5 (as several candidates did) instead of 125.0 is unarguably wrong. Disappointingly, there were few good answers.

Q18.

Also well answered. A common error was to use the wrong sign in part (b).

Q19.

There were few problems in part (a) apart from those candidates who were unable to make the connection between the number of minutes past 8.00am and the actual time. Even these candidates generally recovered in part (b), which was answered well. Part (c) was often omitted. Many of those who attempted it seemed to be looking for a more complex answer than the one expected. Despite the italicised warning in part (d), many candidates made general points instead of relating their answers to the data given in the question.

Q20.

Again most candidates were able to make a good start on this question.

Q21.

Many candidates scored high marks and there were no problems.

Q22.

Part (a) was well answered. In part (b), a substantial number of candidates worked backwards from Table 3. This is more difficult and likely to be less accurate (even if interpolation is used) than using Table 4. There were surprisingly few correct answers to part (b), with the majority of candidates adding (instead of subtracting) 1.2816σ to the mean.

Q23.

This question was generally well answered.

Q24.

This proved an easy starter for all but the small minority who confused standard deviation with variance.

Q25.

This question was well answered with part (a) causing fewer problems than similar past questions. This may have been due to the correct z-value being positive rather than negative.